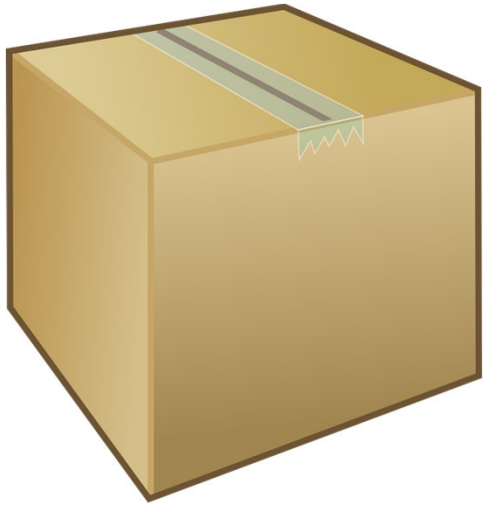
Evaluating Uncertainty Estimation: Calibration

Calibration: intuition

- Calibration is the notion (measure) of statistical consistency between the predictive distributions and the actual observations

We predict it is a 75% chance that there is a cat in such boxes.

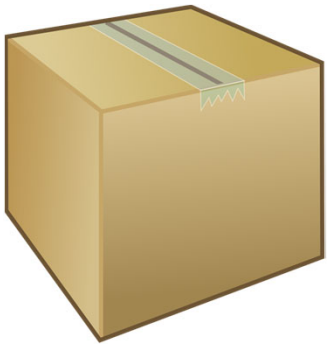
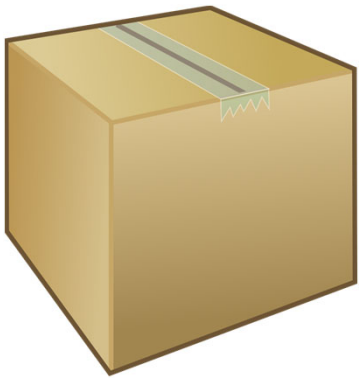
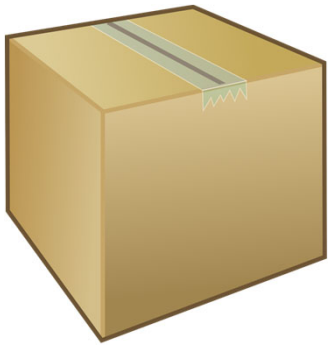
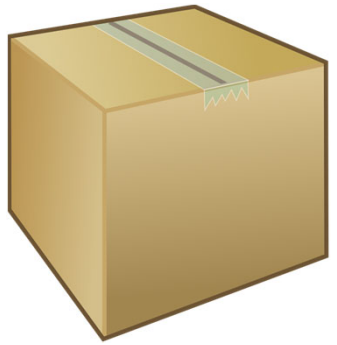


Evaluating Uncertainty Estimation: Calibration

Calibration: intuition

- Calibration is the notion (measure) of statistical consistency between the predictive distributions and the actual observations

We order 4 of such boxes

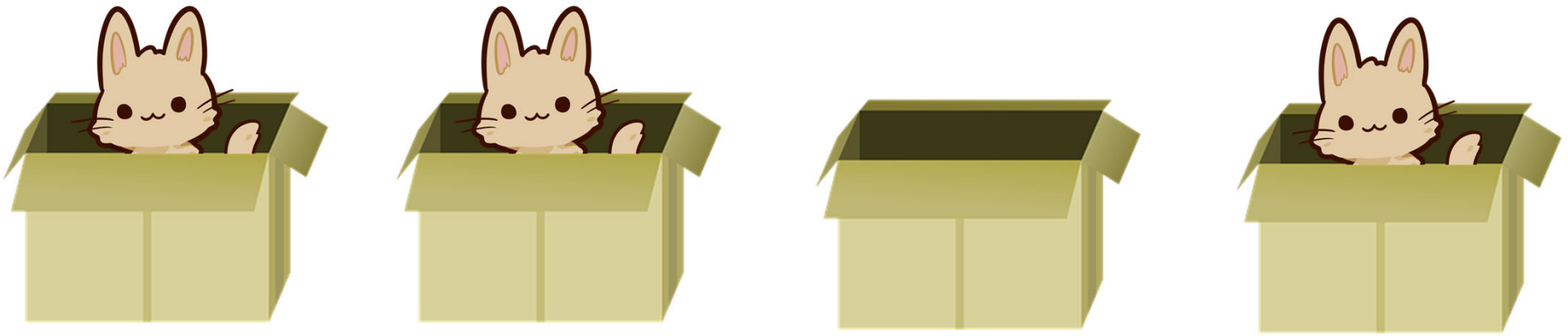





Evaluating Uncertainty Estimation: Calibration

Calibration: intuition

- Calibration is the notion (measure) of statistical consistency between the predictive distributions and the actual observations

And open them to find 3 of 4 boxes actually contain a cat!



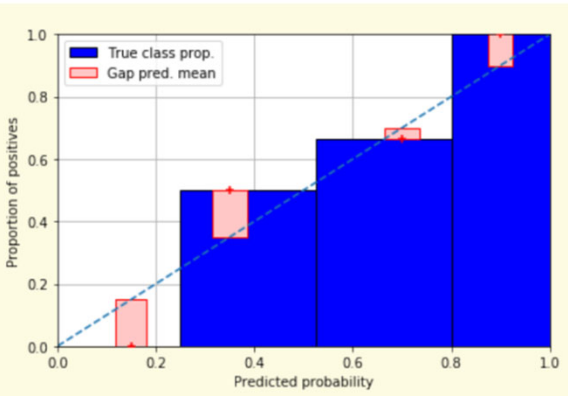
So our predictive distribution for such boxes was calibrated

Evaluating Uncertainty Estimation: Calibration

Calibration: example measures

- Log likelihood
- Expected Calibration Error (ECE)/Reliability Diagram

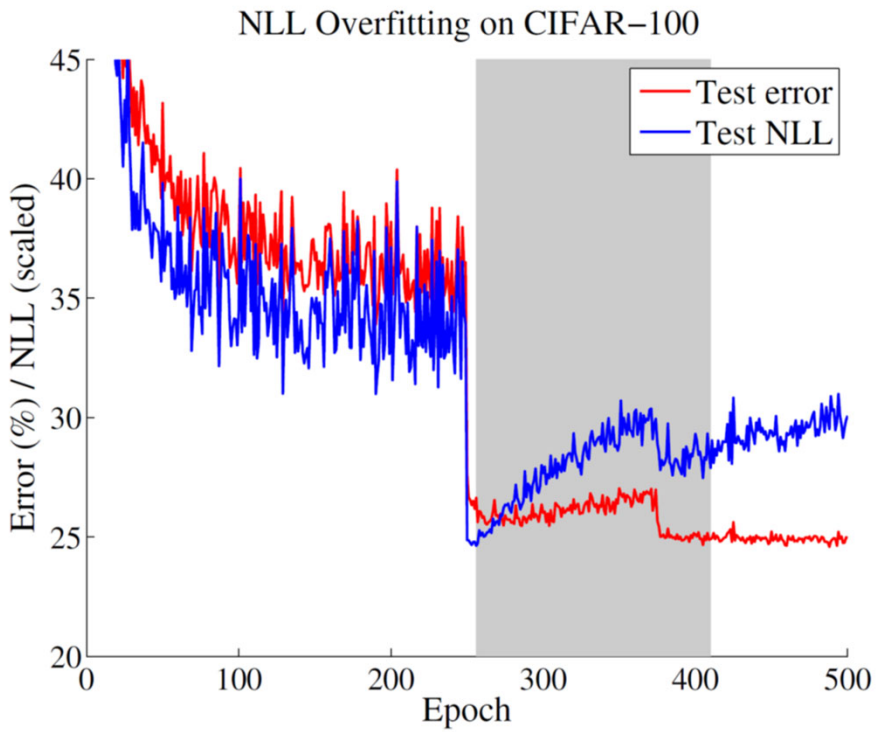
	$\hat{p}$	y
0	0.1	0
1	**0.2**	0
2	**0.3**	0
3	0.4	1
4	**0.6**	0
5	0.7	1
6	**0.8**	1
7	0.9	1

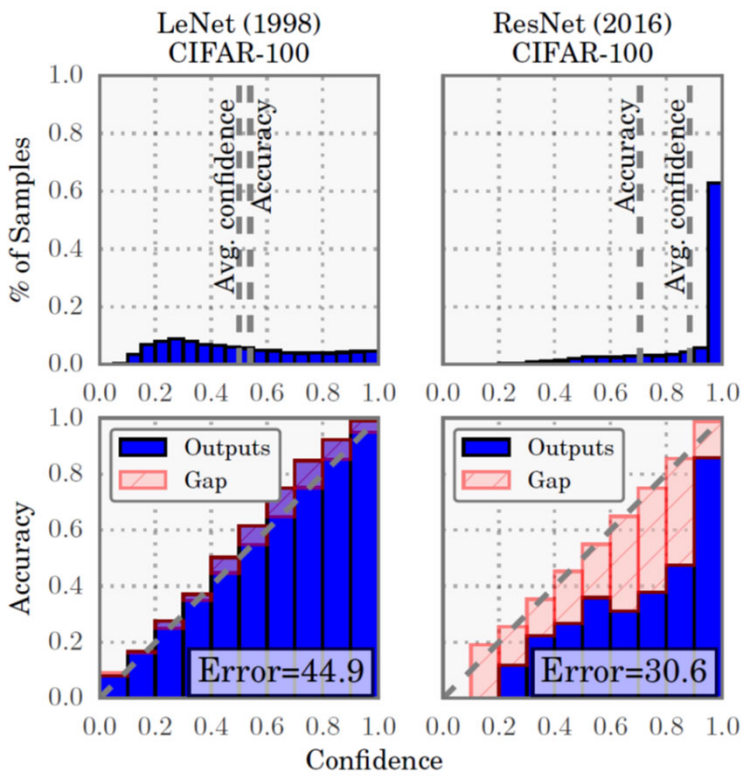


[source: https://classifier-calibration.github.io/]

Evaluating Uncertainty Estimation: Calibration

Calibration: the problem with softmax





["On Calibration of Modern Neural Networks" ICML 2017]

Evaluating Uncertainty Estimation: Calibration

Calibration: methods

$$p(i) = \frac{e^{\frac{f(i)}{T}}}{\sum_j e^{\frac{f(j)}{T}}}$$

- Techniques to calibrate are usually simple but important
- Can be applied as post processing (in many cases)
- Achieve significant improvement on the quality of uncertainty
- Examples
  - Platt Scaling
    - [“Advances in Large-Margin Classifiers”, MIT Press 2000]
  - Beta Calibration
    - [“Beyond Sigmoids: How to obtain well-calibrated probabilities from binary classifiers with beta calibration”, 2017]
  - Dirichlet Calibration
    - [Beyond temperature scaling: Obtaining well-calibrated multiclass probabilities with Dirichlet calibration.” NeurIPS 2019]
  - Isotonic Regression
    - [“PAV and the ROC convex hull.” Machine Learning 2007]
    - [“Obtaining calibrated probability estimates from decision trees and naive Bayesian classifiers” ICML 2001]
  - Other works
    - [“Accurate Uncertainties for Deep Learning Using Calibrated Regression” ICML 2018]
    - ["Bias-Reduced Uncertainty Estimation for Deep Neural Classifiers", ICLR 2019

